DCL-BPSK

BPSK / DPSK / DEPSK

MODULATION / DEMODULATION KIT

EXPERIMENTAL MANUAL

…. A MESSAGE FROM

AKADEMIKA

Today's system designers are faced with tomorrow's problems. DIGITAL COMMUNICATION is one of the important subject need to teach while learning electronics.
It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

OUR MOTTO:
- Light years ahead – refers to leadership.

As leaders in our industry in India, we are totally committed to servicing as the standard against which all are measured in the areas of

• Design • Quality • Value • Delivery • Support.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.
As you will move through this manual you will quickly discover that we have complete, truly innovative & superior training products. We are so committed to quality that we back our products with a complete comprehensive warranty.

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SAFETY RULES

Carefully follow the instructions contained in this manual as they provide you with important points on safety during the installation, use and maintenance. Keep this manual always with you for easy reference.

Arrange all accessories in order after unpacking, so that its integrity is checked with respect to its checklist. Also ensure that no visible damage as such appear on any accessories.

Before connecting the power supply to the kit, be sure that the jumpers and connecting chords are connected correctly as per experiment.

In DCL-BPSK/DPSK/DEPSK kit, signals are to be monitored with an oscilloscope or with help of indicators as explained in the manual.

This kit must be employed only for the use for which it has been conceived, i.e. as educational equipment and must be used under the direct survey of expert personnel. Any other use is improper and so dangerous. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

In case of any fault or malfunctioning in the trainer kit, turn off the power supplyand do not tamper the kit. In case servicing is required, contact the service center for technical assistance.

The kits are liable to malfunction/under-perform if they are not operated underfollowing conditions,

- Ambient temperature: from 0 to 45 ° C.
- Relative humidity: from 20 to 80 %.

Avoid any immediate/significant change of temperature and humidity.

WARRANTY

These kits are warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts .When the failure is result of user's neglect, natural disaster or accident, we charge for repairs regardless of the warranty period. The warranty does not cover perishable items like connecting chords, Crystals, etc. and other imported items.

This warranty is subject to the following conditions and limitations. The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product is, on Akademika Lab Solutions sole judgment. The defective product will be replaced with new one or repaired without charge or with charge.

In the warranty period if the service is needed, purchaser should get in touch with the service center or sales outlet. The purchaser should return the product to the service center or sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of kits has to be mentioned specifically. We return the product to the purchaser at our expense. In case that warranty does not cover the product on Akademika Lab Solutions judgment, we repair the product after obtaining prior permission from purchaser who receives an estimate statement about repairing charges. In such case, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but have no rights to stretch the meaning of original warranty contents or offer additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and up-gradation of products, be sure to contact our service center or sales outlet.

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INTRODUCTION

AKADEMIKA with the trainer, namely, “DCL-BPSK/DPSK/DEPSK” initiates the user to the various Data Conditioning and Carrier Modulation Techniques, normally adopted in practice.

DCL-BPSK/DPSK/DEPSK MODULATION / DEMODULATION KIT

FEATURES:

1) The System illustrates the use of the following data formats:
a) Non Return to Zero - Level (NRZ-L)
b) Differentially encoded NRZ-L

2) The trainers come with an onboard data simulator, which generates the NRZ-L pattern depending on the positions of switches of 8-bit DIP switch and the reference clock (256 KHz), which enable the trainers to work in a standalone mode.

3) Three carrier modulation options are available on the kit.
a) Binary phase shift Keying (BPSK),
b) Differential phase Shift Keying (DPSK),
c) Differentially Encoded Phase Shift Keying (DEPSK).

4) Onboard carrier generation circuit generates sine waves synchronized to the transmitted data. The carrier sine waves generated onboard are of frequencies.
a)1MHz (0 Deg.)
b)1MHz (180 Deg.)

5) Following technique is used for detecting the information from the carrier:
1) Multiplier and FIR filter detection for BPSK/DPSK/DEPSK.

6) Interconnection facilities:
Sockets and connecting chords are provided for connections on board and for connections.

7) Test Points:
All relevant test points are brought-out for observations. Observations are carried out on an oscilloscope.

TECHNICAL SPECIFICATIONS

DCL- BPSK/DPSK/DEPSK MODULATION AND DEMODULATION KIT

DATA SIMULATOR	**:**	Onboard 8-bit variable NRZ-L pattern.
CRYSTAL OSCILLATOR	**:**	32.768 MHz
CLOCK FREQUENCY	**:**	256 KHz.
DATA ENCODING	**:**	NRZ-L, Differentially encoded NRZ-L.
ONBOARD CRRIER SINE WAVES	**:**	1MHz (0°) & 1MHz (180°).
CARRIER MODULATION	**:**	BPSK, DPSK,DEPSK.
CARRIER DEMODULATION	**:**	BPSK, DPSK,DEPSK.
DATA DECODING	**:**	NRZ-L, Differentially encoded NRZ-L.
INTERMEDIATE SIGNAL	**:**	Provision for observing intermediate signal during demodulation.
INTER CONNECTION	**:**	2 mm banana socket.
POWER SUPPLY	**:**	+12V, -12V, +5V, GND.
TEST POINTS	**:**	18

ACCESSORIES

	DCL-BPSK	Quantity
1	Short Links PCS2 Patch cord Male To Male	10
2	ADCL-01 EXPT Manual	1
3	5 PIN DIN CONNECTOR Power Supply	1
4	Power cord	1

PRINCIPLES OF ADVANCED DIGITAL MODULATION AND DEMODULATION TECHNIQUES

The Digital data in the systems can be encoded in several formats. All these waveforms can be broadly classified into the following four groups:
1. Non-return to Zero formats,
2. Return to Zero formats,
3. Phase encoded formats,
4. Multilevel binary formats.

It is common that some puzzling questions may arise to a beginner as to why there should be so many PCM Waveforms? And are there really so many unique applications necessitating such a variety of waveforms to represent the "ones" and"zeros"? The reason for the large selection relates to the difference in performances that characterizes each waveform. In choosing a coding scheme for a particular application, the common parameters worth examination are as follows:

1) DC Component:
The DC component associated with the data formats has to be minimized, if the system is to be AC coupled. The proper selection of the data format ensures optimum DC component that is associated with the signal power spectrum.

2) Self Clocking:
The presence of sufficient number of transition in the transmitted data enables the receiver to derive the clock from the data. Thus it is preferable to select a data format, which enables sufficient number of transitions in the transmitted data stream. For example, the biphase signals allow sufficient number of transitions in the transmitted data, compared to the corresponding Non-return to zero signals.

3) Band Width:
The proper selection of the coding enables to optimize the bandwidth requirement.

4) Noise Immunity:
The various PCM waveform types can be further characterized by the probability of bit error versus signal to noise ratio. For Example, the NRZ waveforms have better immunity to noise than the corresponding unipolar return to zero signal, which enables an error free transmission.

FUNDAMENTALS OF CODING AND CARRIER MODULATION TECHNIQUES:

Digital Communication Systems represent the information by the binary digits, 'Ones` and 'Zeros`. The basic principles of digital communication are Sampling, Multiplexing, Encoding, Error Control Coding, Data Conditioning, and Carrier Modulation etc. In coding we represent sample data pattern in different formats. In carrier modulation, carrier is modulated as per modulating signal.

CARRIER MODULATION SCHEMES:

Digital Communication is the technique by which the whole information is represented in terms of binary digits, i.e., ‘ones’ and ‘zeros’. These digits are represented by discrete voltage levels and the clock frequency of a digital communication scheme is generally low. For long distance transmission, the data is made to modulate a continuous wave (sine wave) carrier. These techniques are called Carrier Modulation Techniques. The various types of Carrier Modulation Techniques normally adopted in practice fall under three broad categories.

a) Amplitude Shift Keying (ASK)
b) Frequency Shift Keying (FSK)
c) Phase Shift Keying (PSK)

a) AMPLITUDE SHIFT KEYING (ASK):
For all types of Carrier Modulation, the Carrier frequency should be at least 2times base band of the modulating signal.
Assuming that the modulated carrier is a sine wave represented by the following equation.
C (t) = A (t) • cos (wt)
Where C (t) = Carrier sine wave.
A (t) = Time varying amplitude
wt = Time varying angle.
Now in amplitude shift keying, the carrier is being transmitted only when the modulating data is ‘one’ and when the data is ‘zero’ the carrier is rejected from transmission. Thus the resulting modulated output of this type of modulation can be represented as follows:
M (t) = r (t) • C (t)
Where M (t) = Modulated Carrier
r (t) = Time varying modulating data which is either ‘one’ or ‘zero’
Now,
M (t) = A (t) • cos (wt) when modulating data is one.
= 0 when modulating data is zero.
An Envelope detector is used to recover the data from the modulated carrier.

b) FREQUENCY SHIFT KEYING (FSK):

In this type of modulation, the modulated output shifts between two frequencies for all ‘one’ to ‘zero’ transitions.
Let the carrier frequencies be represented by w1 and w2, and then we have:
M (t) = A (t) • Cos (w1t) if data is ‘One’
A (t) • Cos (w2t) if data is ‘Zero’.
Where A (t) = Time varying amplitude of the sine wave.
M (t) = Modulated carrier.
FSK Demodulator employs PLL logic for the recovery of data.

c) PHASE SHIFT KEYING (PSK):
In the PSK modulation or phase shift keying, for all ‘one’ to ‘zero’ transitions of the modulating data, the modulated output switches between the in phase and out of phase components of the modulating frequency. If the modulated carrier is represented by:

M (t)	=	A (t) • Cos (wt + Phase)
Where A (t)	=	Time varying amplitude, and
wt	=	Time varying angle.
M (t)	=	Modulated carrier.

Then the phase equals 0° whenever the data equals ‘one’ and the phase equals180° whenever the data equals ‘zero’. This type of phase shift keying is called Binary phase Shift Keying (BPSK).

There are also other forms of phase shift keying like Differential Phase Shift Keying (DPSK), Differentially Encoded Phase Shift Keying (DEPSK) Quadrature Phase Shift Keying (QPSK), and Differential Quadrature Phase Shift Keying (DQPSK) etc.

In DPSK/DEPSK the data stream to be transmitted is encoded in such an away to get the differentially encoded data as input to DPSK/DEPSK modulator. Then the phase equals 0° whenever the data equals ‘one’ and the phase equals 180° whenever the data equals ‘zero’. This type of phase shift keying is called Differential phase Shift Keying (DPSK)/ differentially encoded phase Shift Keying (DEPSK).

In QPSK, phase has 4 different values like 0°, 90°, 180° and 270°. They are used in the carrier modulation of differentially coded debit pair. One of the four carriers is transmitted at a time with respect to the symbol generated by EVEN&ODD.

In DQPSK the inputs to the modulator (EVEN&ODD) are differentially encoded before applying to the modulator & then transmitted similar to the QPSK. At the receiver recovered data decoded before applying to the data decoder.

The BPSK/DPSK/DEPSK detector works on the principle of square law. BPSK/DPSK/DEPSK modulated carrier can be represented mathematically as:

M (t)	=	A (t) • Cos (wt + Phase)	
We have phase	=	0°	- when data is 1
	=	180°	- when data is 0
Thus,			
M (t)	=	A (t) • Cos (wt)	- when data is 1
	=	-A (t) • Cos (wt)	- when data is 0

On squaring the modulated carrier, the negative sign gets eliminated and the frequency gets multiplied by two.

M2 (t)	=	A2 (t) • cos2 (wt)	- for both data is 1 and 0
M2 (t)	=	A2 (t) • (1+cos (2wt))/2	

Now the in phase reference carrier can be recovered by dividing the frequency of the squared modulated carrier by two.

Once the carrier is recovered, the data can be detected by comparing the phase of the received modulated carrier with the phase of the reference carrier.

EXPERIMENT
NO. 1

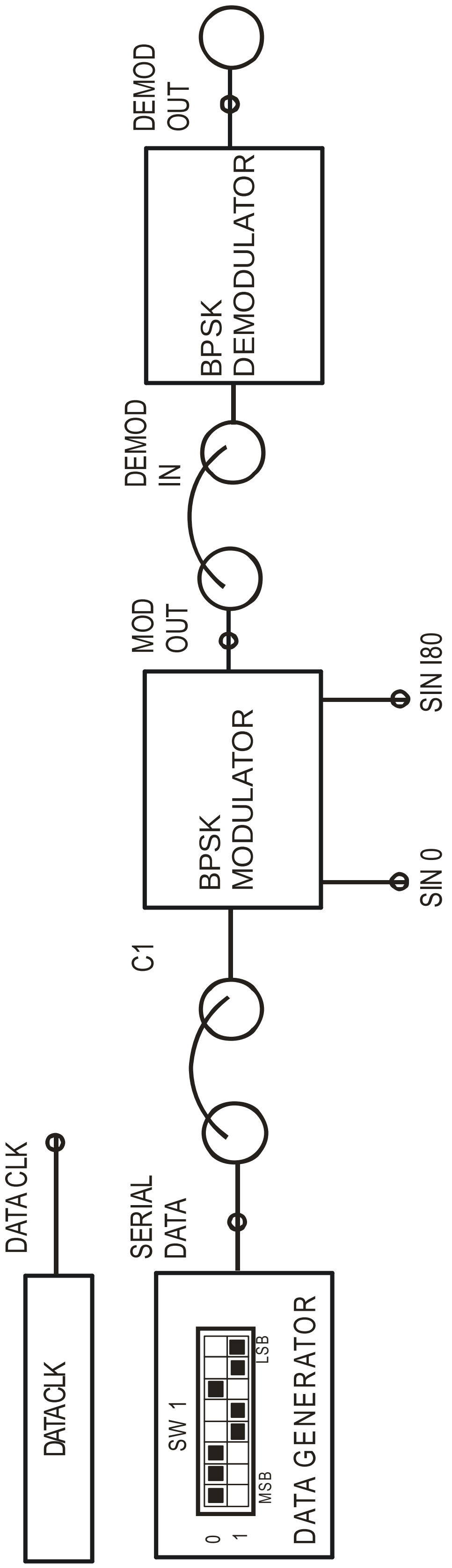

FIG 1.1 BLOCK DIAGRAM FOR
BINARY PHASE SHIFT KEYING MODULATION TECHNIQUE

EXPERIMENT NO: 1

NAME:

BINARY PHASE SHIFT KEYING MODULATION TECHNIQUES

OBJECTIVE:

Study of Carrier Modulation Techniques by Binary Phase Shift Keying method

THEORY:

In BPSK, individual data bits are used to control the phase of the carrier. During each bit interval, the modulator shifts the carrier to one of two possible phases, which are 180 degrees or π radians apart. This can be accomplished very simply by using a bipolar baseband signal to modulate the carrier's amplitude, as shown in Figure 1.1.The output of such a modulator can be represented mathematically as

$$x(t) = R(t) \cos(\omega_c t + \theta) \qquad \text{--Equation 1}$$

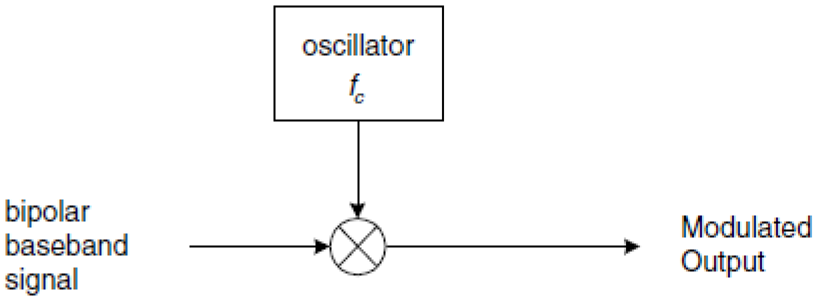

FIG. 1.2 BPSK MODULATOR

Where R(t)is the bipolar baseband signal, ω_c is the carrier frequency, and θ is the phase of the unmodulated carrier. If the output of the modulator is to be represented in complex-envelope form referenced to the carrier frequency, the modulated signal is given as

$$x(t) = I(t) + jQ(t) \qquad \text{--Equation 2}$$

Where

$$I(t) = R(t) \cos\theta$$

$$Q(t) = R(t) \sin\theta$$

In the special case of θ = 0, Equation 2 reduces to

$$\tilde{x}(t) = R(t)$$

And the real-valued baseband signal can be used directly as the complex-envelope representation of the modulator output. However, to allow for subsequent phase shifting, the signal's complex-envelope representation should always be implemented as a complex-valued signal. For the special case of $\theta = 0$, the imaginary part of the complex signal is simply set to zero.

BPSK DEMODULATION:

A correlation receiver for BPSK is shown in Figure 1.2. The modulated signal is multiplied by the recovered carrier, and this product is integrated over a bit interval. If the integration result is positive, the received bit is considered to be 1; if the integration result is negative, the received bit is considered to be 0.

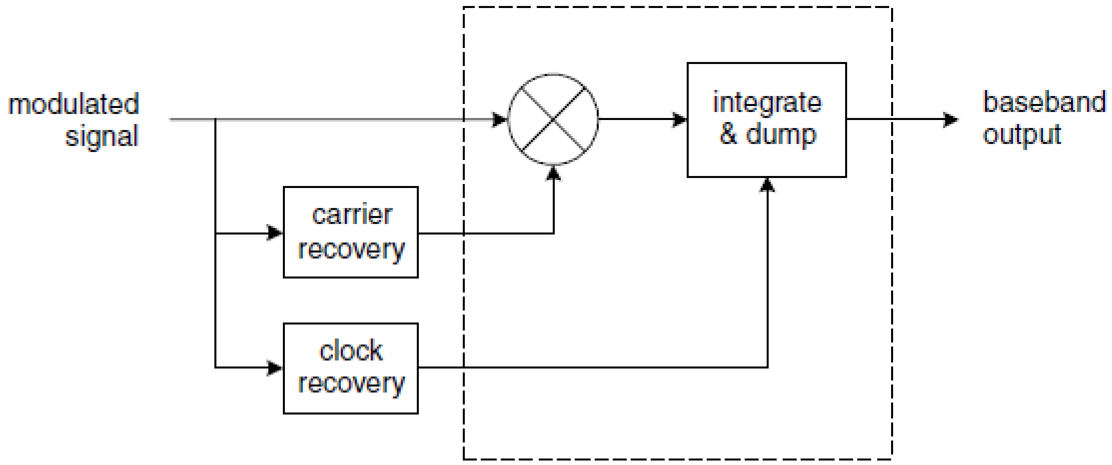

FIG.1.3 CORELATION RECEIVER FOR BPSK

<u>EQUIPMENTS</u>:

- DCL-BPSK Kit
- Connecting Chords
- Power supply
- DSO

<u>PROCEDURE</u>:

1. Refer to the block diagram (Fig.1.1) and carry out the following connections.
2. Connect power supply in proper polarity to the kit **DCL-BPSK** and switch it on.
3. Select Data pattern of simulated data using switch **SW1**.
4. Connect **SERIAL DATA** generated in **DATA GENERATOR** to **CI** of **BPSK MODULATOR**.
5. Observe the BPSK modulated data at **MOD OUT** post as shown in (fig 1.2).
6. Connect the **MOD OUT** of **BPSK MODULATOR** to the **DEMOD IN** of **BPSKDEMODULATOR**.
7. Observe the BPSK demodulated data at the output of the **BPSKDEMODULATOR** at **DEMOD OUT.**
9. Observe various waveforms as mentioned below (Fig. 1.2).

OBSERVATION:

Observe the following waveforms on oscilloscope and plot it on the paper.

1. Transmitter clock **DATA CLOCK** (Fig. 1.4 (a))
2. NRZ-L coded data **SERIAL DATA** (Fig. 1.4 (a))
3. Carrier frequency **SIN 0** and **SIN 180** (Fig. 1.4 (b))
4. **MOD OUT** with respect to **SERIAL DATA** (Fig. 1.4 (c))
5. **MID OUT** with respect to **SERIAL DATA** (Fig. 1.4 (d))
6. **DEMOD OUT** with respect to **SERIAL DATA** (Fig. 1.4 (e))

CAPTURED WAVEFORM:

CH 1 : DATA CLK (256 KHz) & CH 2 : SERIAL DATA

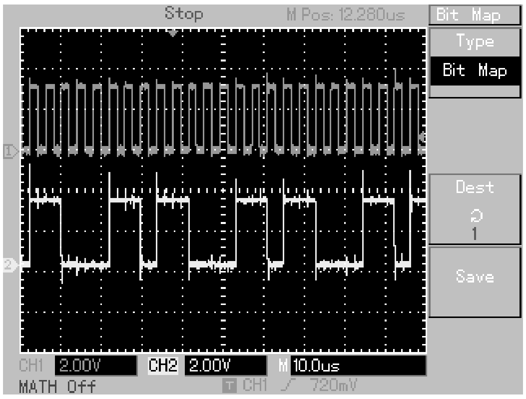

FIG. 1.4(a)

CH 1 :SIN 0 (1MHz) & CH 2 : SIN 180 (1MHz)

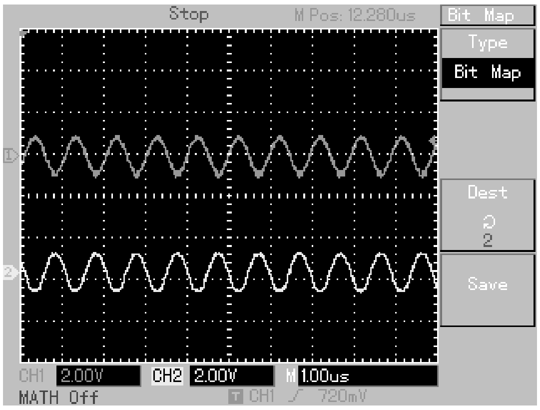

FIG. 1.4(b)

CH 1 :SERIAL DATA & CH 2 : MOD OUT

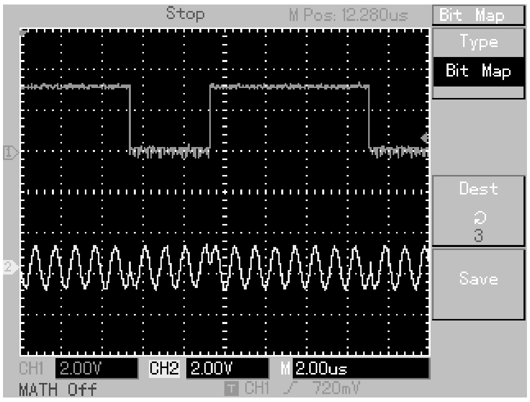

FIG. 1.4(c)

CH 1 :SERIAL DATA&CH 2 : MID OUT

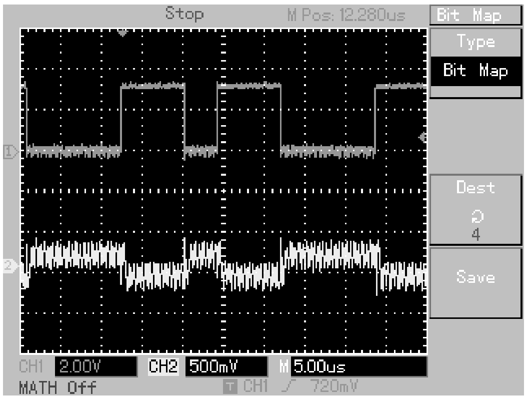

FIG. 1.4(d)

CH 1 :SERIAL DATA& CH 2 : DEMOD OUT

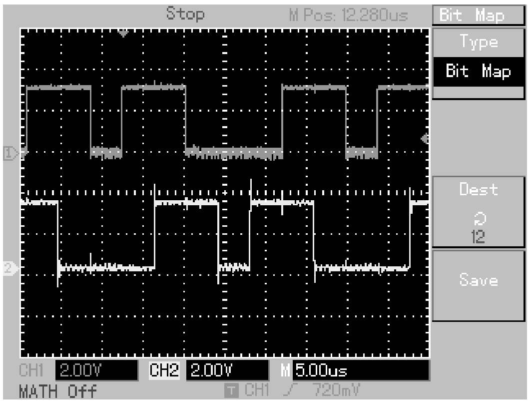

FIG. 1.4(e)

CONCLUSION:

It is observed that the successful operation of the BPSK detector is fully dependent on the phase components of the transmitted modulated carrier. If the phase reversals of the modulated carrier along with the rising and falling edges of the data are not proper, then the efficient detection of data from BPSK modulated carrier becomes impossible.

EXPERIMENT

NO. 2

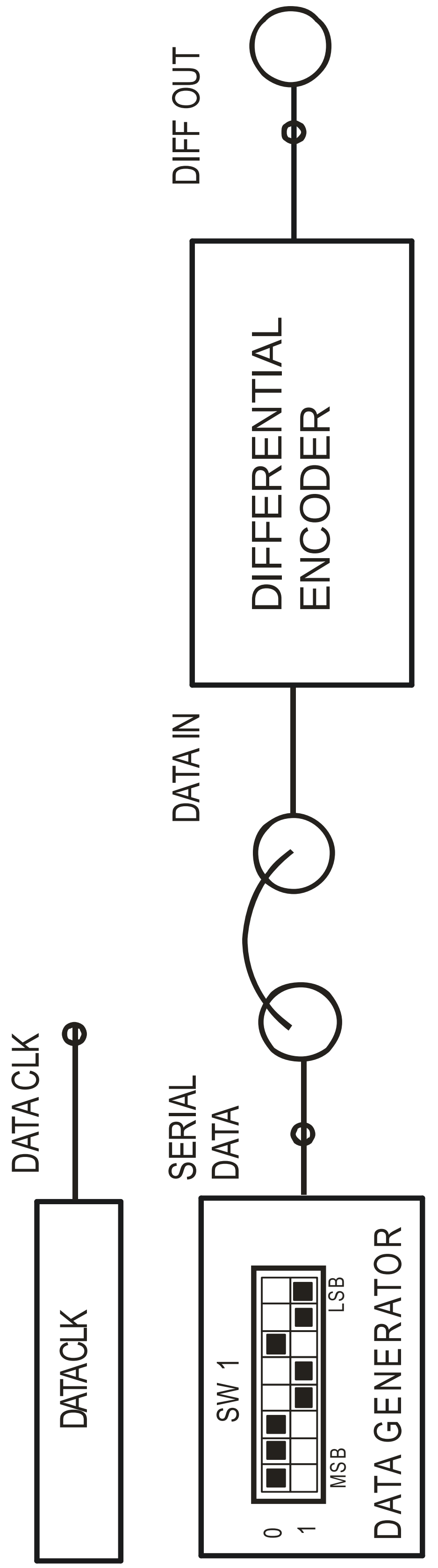

FIG 2.1 BLOCK DIAGRAM FOR
DIFFERENTIAL ENCODING OF NRZ-L DATA

EXPERIMENT NO: 2

NAME:

DIFFERENTIAL ENCODING OF DIBIT DATA

OBJECTIVE:

To study differential encoding techniques for debit data

THEORY:

Differential encoding logic:

A means of generating a differentially encoded data is shown in fig.2.2. The data stream to be transmitted, d (t), is applied to one input of an exclusive-OR logic gate. To the other gate input is applied the output of the exclusive-OR gate b(t) delayed by the time Tb allocated to one bit. This second input is then b (t-Tb).

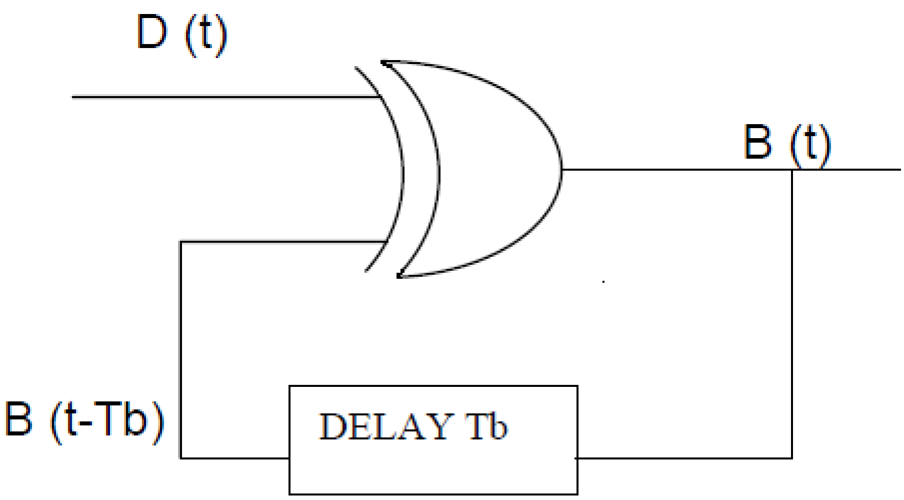

FIG. 2.2 DIFFERENTRIAL ENCODER

In fig, 2.3 we have drawn the logic waveform to illustrate the response b(t) to an input d(t). The upper level of the waveform corresponds to logic 1,the lower level to logic 0. Because of the feedback involved in the system of fig 2.3 there is difficulty in determining the logic levels in the interval in which we start to draw the Waveform (interval 1 in fig. 2.3) we cannot determine b (t) in this first interval of our waveform unless we know b (k=0). But we cannot determine b(0) unless we know both d(0) and b(-1), etc. thus, to justify any set of logic levels in an initial bit interval we need to know the logic levels in the preceding interval. But such a determination requires information about the interval two bit times earlier and so on. In the waveform of fig.2.3 we have circumvented the problem by arbitrarily assuming that in the first interval b (0)=0. It is shown below that, the data will be correctly determined regardless of our assumption concerning b(0).

We now observe that the response of b(t) to d(t) is that b(t) changes level at the beginning of each interval in which d(t)=1 and b(t) does not changes level when d(t)=0. Thus during interval 3, d(3)=1, and correspondingly b(3) changes at the beginning at that interval. During interval 6 & 7, d(6) = d(7) =1 and there are changes in b(t) at the beginning of both intervals. During bits 10,11,12,and 13d(t)=1 and there are changes in b(t) at the beginnings of each of these intervals This behavior is to be anticipated from the truth tables of the exclusive- OR gate. For we note that when d(t)=0, b(t)=b(t-Tb) so that, whatever the initial value of b(t-Tb), it reproduces itself. On the other hand when d (t)=1 then b(t)=/b(t-Tb). Thus, in each successive bit intervals b(t) changes from its value in the previous interval.

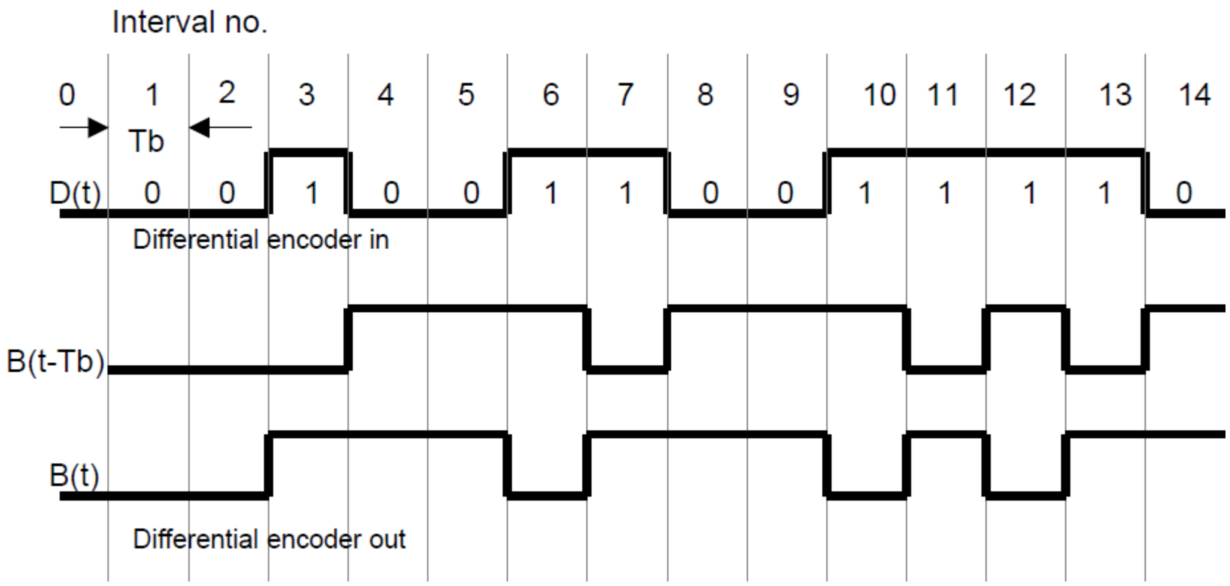

FIG. 2.3 WAVEFORM OF DIFFERENTIAL ENCODING OF DATA

Note that in some intervals where d(t)=0 we have b(t)=0 and in other intervals when d(t)=0 we have b(t)=1. Similarly, when d(t)=1 sometimes b(t)=1 and sometimes b(t)=0. Thus there is no correspondence between the levels of d(t) and b(t), and the only invariant feature of the system is that a change (sometime up and sometime down) in b(t) occurs whenever d(t)=1, and that no change in b(t) will occur whenever d(t0)=0.
Finally, we note that in waveform of fig.2.3 are drawn on the assumption that, in interval 1, b(0)=0. As is easily verified, if not intuitively apparent, if we had assumed b(0)=1, the invariant feature by which we have characterized the system would continue to apply. Since b(0) must be either b(0)=0 or b(0)=1, there being no other possibilities, our result is valid quite generally. If, however, we had started with b(0)=1 the levels b(1) and b(0) would have been inverted.

<u>EQUIPMENTS</u>:

- DCL-BPSK Kit
- Connecting Chords
- Power supply
- DSO

PROCEDURE:

1. Refer to the block diagram (Fig.2.1) and carry out the following connections.
2. Connect power supply in proper polarity to the kit **DCL-BPSK** and switch it on.
3. Select Data pattern of simulated data using switch **SW1**.
4. Connect **SERIAL DATA** generated at the **DATA GENERATOR** to **DATA IN** of **DIFFERENTIAL ENCODER**.
5. Connect the coded data **NRZ-L DATA** to the **DATA IN** of the **DIFFERENTIAL ENCODER.**
6. Observe the input data and the differentially encoded data on test points **SERIAL DATA** and **DIFF OUT** respectively.

OBSERVATION:

ON KIT **DCL-BPSK** observe the following waveforms on oscilloscope and plot it on the paper.

1. Differentially encoded data **DIFFOUT** with respect to **SERIAL DATA** (Fig.2.4 (a))

CAPTURED WAVEFORM:

CH 1 : SERIAL DATA & CH 2 : DIFF OUT

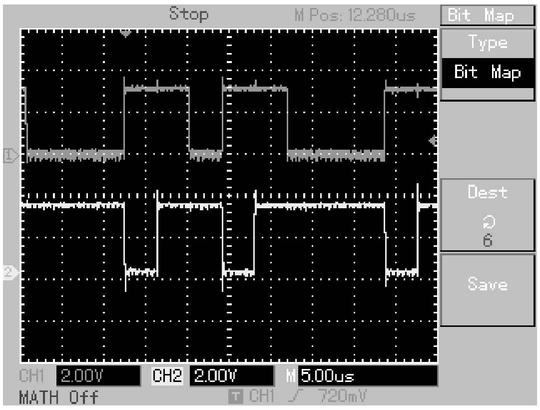

Fig.2.4 (a)

CONCLUSION:

It is observed that when input data to the differential encoder contains “1” the output of the differential encoder makes transition from its previous state. If input data contains “0” then output of differential encoder maintains the previous state.

EXPERIMENT

NO. 3

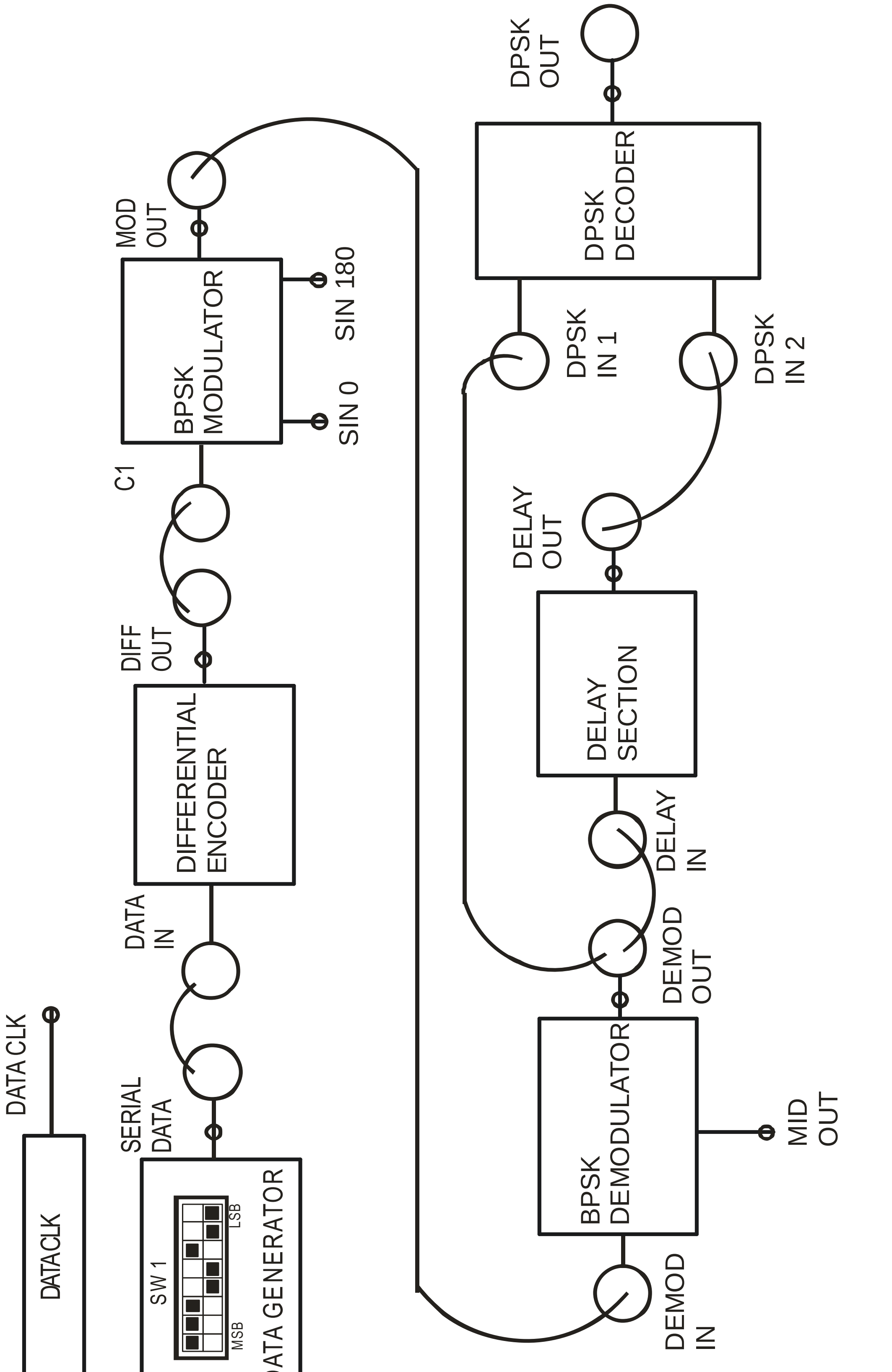

FIG 3.1 BLOCK DIAGRAM FOR DIFFERENTIAL PHASE SHIFT KEYING MODULATION TECHNIQUE

EXPERIMENT NO: 3

NAME:

DIFFERENTIAL PHASE SHIFT KEYING MODULATION TECHNIQUES

OBJECTIVE:

Study of Carrier Modulation Techniques by differential phase Shift Keying (DPSK) method

THEORY:

In BPSK communication system, the demodulation is made by comparing the instant phase of the BPSK signal to an absolute reference phase locally generated in the receiver. The modulation is called in this case BPSK absolute. The greatest difficulty of these systems lies in the need to keep the phase of the regenerated carrier always constant. This problem is solved with the PSK differential modulation, as the information is not contained in the absolute phase of the modulated carrier but in the phase difference between two next modulation intervals.

Fig.3.2 a & b shows the block diagram of DPSK modulation and demodulation system. The coding is obtained by comparing the output of an EX-OR, delayed of a bit interval, with the current data bits (for detailed explanation see experimentno.2). As total result of operation, the DPSK signal across the output of the modulator contains 180 deg. phase variation at each data bit “1”. The demodulation is made by a normal BPSK demodulator, followed by a DPSK DECODER which is nothing but a decision device supplying a bit “1” each time there is a variation of the logic level across its input.

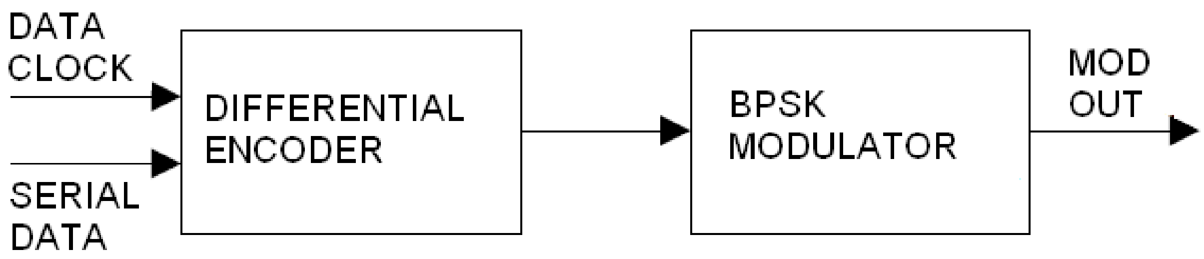

Fig. 3.2 (a) DPSK MODULATOR

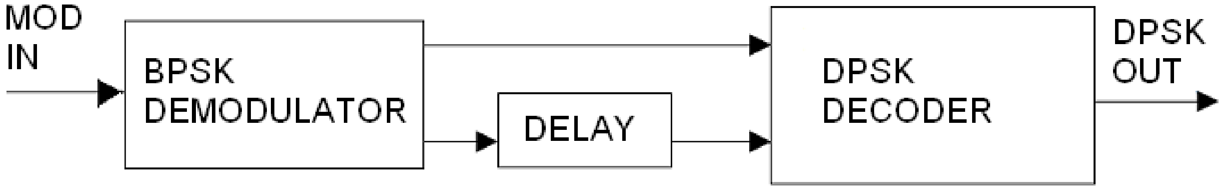

Fig. 3.2 (b) DPSK DEMODULATOR

The DPSK system explained above has a clear advantage over the BPSK system in that the former avoids the need for complicated circuitry used to generate a local carrier at the receiver. To see the relative disadvantage of DPSK in comparison with PSK, consider that during some bit interval the received signal is so contaminated by noise that in a PSK system an error would be made in the determination of whether the transmitted bit was a 1 or 0. In DPSK a bit determination is made on the basis of the signal received in two successive bit intervals. Hence noise in one bit interval may cause errors to two-bit determination. The error rate in DPSK is therefore greater than in PSK, and, as a matter of fact, there is a tendency for bit errors to occur in pairs. It is not inevitable however those errors occur in pairs. Single errors are still possible

EQUIPMENTS:

- DCL-BPSK Kit
- Connecting Chords
- Power supply
- DSO

PROCEDURE:

1. Refer to the block diagram (Fig.3.1) and carry out the following connections.
2. Connect power supply in proper polarity to the kit **DCL-BPSK** and switch it on.
3. Select Data pattern of simulated data using switch **SW1**.
4. Connect the **SERIAL DATA** to the **DATA IN** of the **DIFFERENTIALENCODER.**
5. Connect differentially encoded data **DIFF OUT** of **DIFFERENTIAL ENCODER** to control input **C1** of **BPSK MODULATOR.**
6. Connect DPSK modulated signal **MOD OUT** of **BPSK MODULATOR** to **DEMOD IN** of the **BPSK DEMODULATOR**.
7. Connect **DEMOD OUT** of **BPSK DEMODULATOR** to **DELAY IN** of **DELAY SECTION** and one input **DPSK IN 1** of **DPSK DECODER**.
8. Connect the **DELAY OUT** of **DELAY SECTION** to the input **DPSK IN 2** of **DPSK DECODER**.
9. Compare the DPSK decoded data at **DPSK OUT** with respect to input **SERIAL DATA.**
10. Observe various waveforms as mentioned below (Fig. 3.3)**.**

OBSERVATION:

ON KIT **DCL-BPSK**

Observe the following waveforms on CRO and plot it on the paper.

1. **SERIAL DATA** with respect to **DATA CLOCK** (Fig.3.3 (a))
2. Differentially encoded **DIFF OUT** with respect to **SERIAL DATA** (Fig.3.3 (b))
3. Carrier frequency **SIN 0** and **SIN 180** (Fig.3.3 (c))
4. **MODOUT** with respect to **DIFF OUT** (Fig.3.3 (d))
5. **MIDOUT** with respect to **DIFF OUT** (Fig.3.3 (e))
6. **DEMODOUT** with respect to **DIFF OUT** (Fig.3.3 (f))
7. **DELAYOUT** with respect to **DEMODOUT** (Fig.3.3 (g))
8. **DPSK OUT** with respect to **SERIAL DATA** (Fig.3.3 (h))

CAPTURED WAVEFORM:

CH 1: DATA CLK (256 KHz) & CH 2: SERIAL DATA

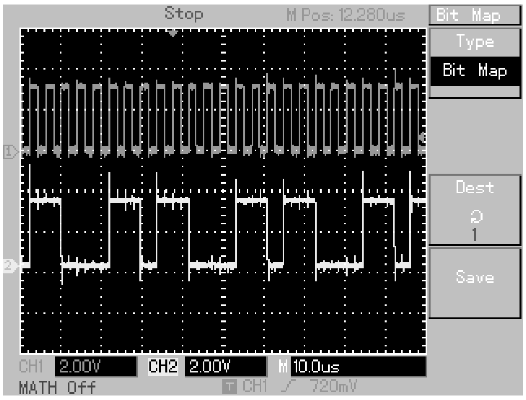

Fig.3.3 (a)

CH 1: SERIAL DATA& CH 2: DIFF OUT

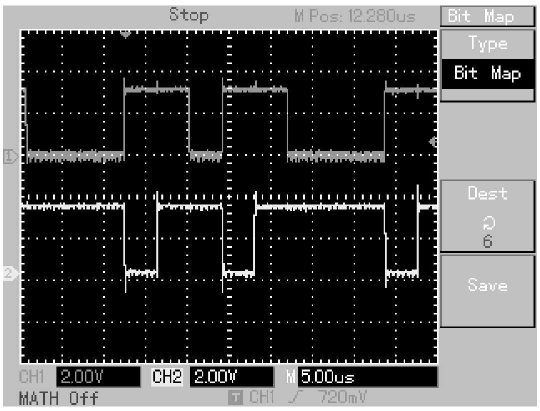

Fig 3.3(b)

CH 1: SIN 0 (1 MHz) & CH 2 : SIN 180 (1 MHz)

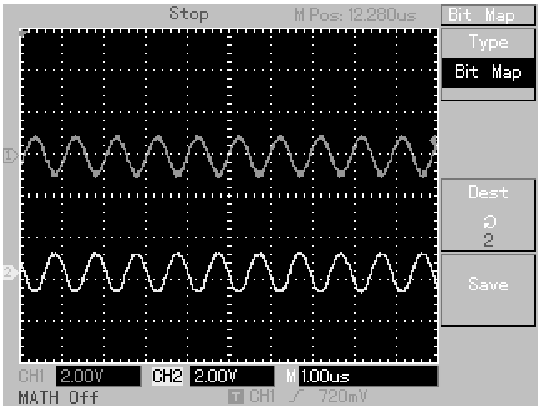

Fig. 3.3 (c)

CH 1: DIFF OUT & CH 2: MOD OUT

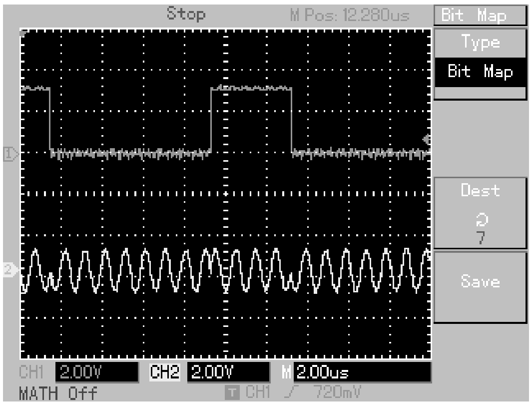

Fig. 3.3 (d)

CH 1: DIFF OUT & CH 2: MID OUT

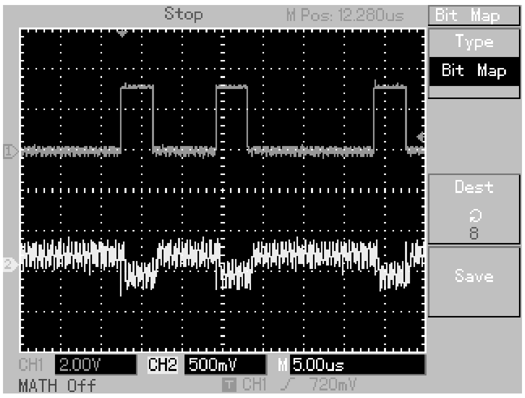

Fig. 3.3 (e)

CH 1: DIFF OUT & CH 2: DEMOD OUT

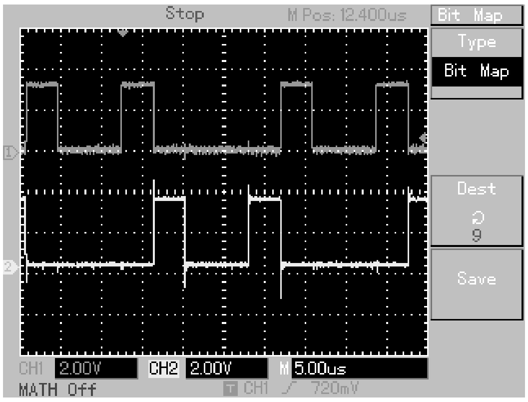

Fig. 3.3 (f)

CH 1: DEMOD OUT & CH 2: DELAY OUT

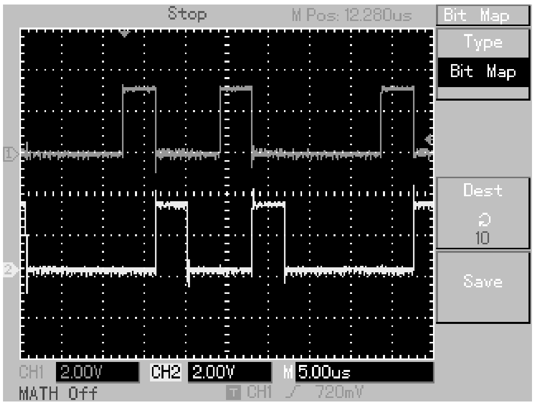

Fig. 3.3 (g)

CH 1: DPSK OUT & CH 2: SERIAL DATA

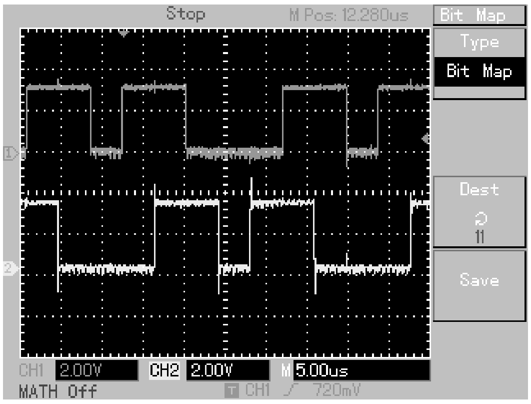

Fig. 3.3 (h)

CONCLUSION:

The differential coding of data to be transmitted makes the bit “1” to be transformed into carrier phase variation. In this way the receiver recognizes one bit “1” at a time which detects a phase shift of the modulated carrier, independently from its absolute phase. In this way the BPSK modulation, which can take to the inversion of the demodulated data, is overcome.

EXPERIMENT NO. 4

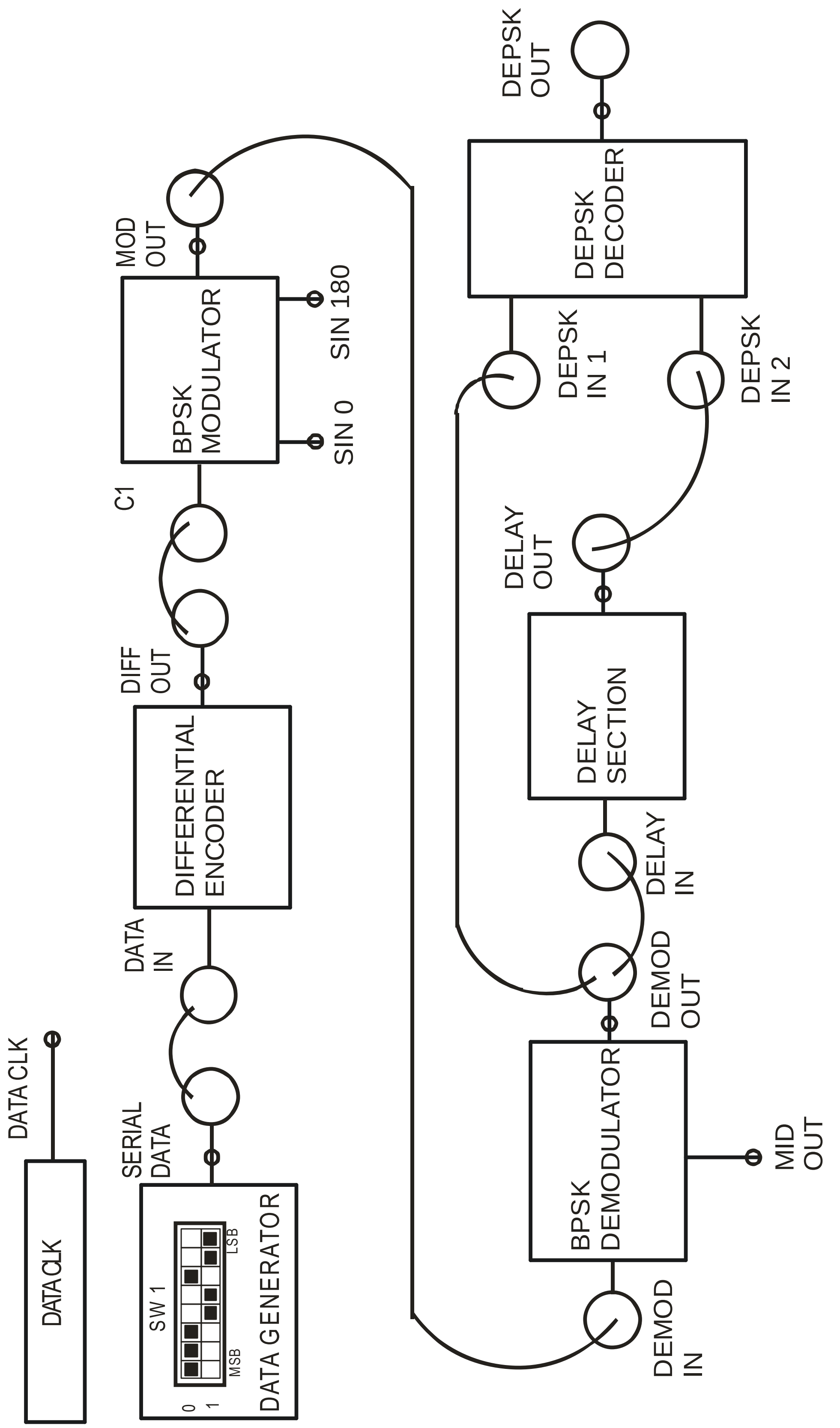

FIG 4.1 BLOCK DIAGRAM FOR DIFFERENTIAL ENCODING PHASE SHIFT KEYING MODULATION TECHNIQUE

EXPERIMENT NO: 4

NAME:

DIFFERENTIAL ENCODING PHASE SHIFT KEYING MODULATIONTECHNIQUES.

OBJECTIVE:

Study of Carrier Modulation Techniques by differentially encoded phase shift keying (DEPSK) method.

THEORY:

The DPSK demodulator requires a device, which operates at the carrier frequency and provides a delay of Tb. Differentially Encoded PSK eliminates the need for such a piece of hardware. In this system, synchronous demodulation recovers the signal b (t), to generate d (t).The transmitter of DEPSK system is identical to the transmitter of the DPSK system as explained in experiment no.3. The signal b (t) is recovered in exactly the same manner for a BPSK system. The recovered signal is then applied directly to one input of an EX-OR logic gate and to the other input is applied b (t-Tb). The gate output will be at one or the other of its level depending on whether b (t)= b (t-Tb) or b (t)=/b (t-Tb). In the first case b (t) did not change level and therefore the transmitted bit is d (t) =0. in the second case d (t)=1.

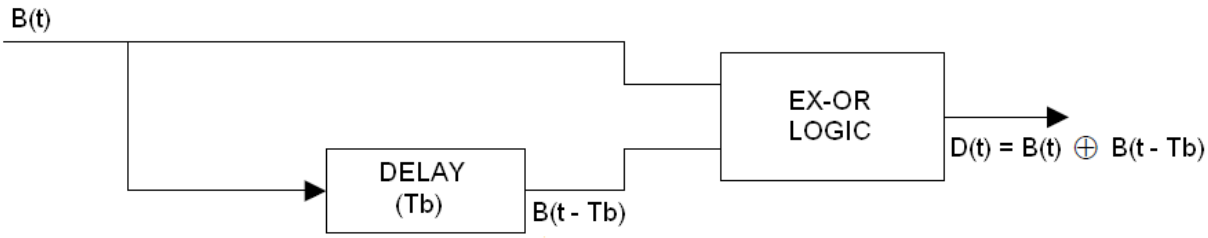

FIG.4.2 DEPSK DECODER

We have seen that in DPSK there is a tendency for bit errors to occur in pairs but that single bit errors are possible. In DEPSK errors always occurs in pairs. The reason for the difference is that in DPSK we do not make hard decision, in each bit interval about the phase of the received signal. We simply allow the received signal in one interval to compare itself with the signal in an adjoining interval and, as we have seen, a single error is not precluded. In DEPSK, a firm definite hard decision is made in each interval about the value of b(t). if we make a mistake , then errors must results from a comparison with the preceding and succeeding bit. This result is illustrated in fig. 4.3. In the fig. 4.3A is shown the error free signals b(k), b(k-1) and d(k)= B(k)⊕B(k-1).In fig. 4.3B we have assumed that b′(k) has a single error. Then b′(k-1) must also have a single error. We note that the reconstructed waveform d′(k) now has two errors.

B(k) : 0 1 1 0 1 1 0 0

B(k-1) : 0 1 1 0 1 1 0 0

$d(k) = B(k) \oplus B(k-1)$: 1 0 1 1 0 1 0

One error
(0 changes to 1)

FIG. 4.3A

b′(k) : 0 1 1 1 1 1 0 0

b′(k-1) : 0 1 1 1 1 1 0 0

$d'(k) = b'(k) \oplus b'(k-1)$: 1 0 0 0 0 1 0

Two errors

FIG.4.3 B

EQUIPMENTS:

- DCL-BPSK Kit
- Connecting Chords
- Power supply
- DSO

PROCEDURE:

1. Refer to the block diagram (Fig.4.1) and carry out the following connections.
2. Connect power supply in proper polarity to the kit **DCL-BPSK** and switch it on.
3. Select Data pattern of simulated data using switch **SW1**.
4. Connect the **SERIAL DATA** to the **DATA IN** of the **DIFFERENTIALENCODER.**
5. Connect differentially encoded data **DIFF OUT** of **DIFFERENTIAL ENCODER** to control input **C1** of **BPSK MODULATOR.**
6. Connect DPSK modulated signal **MOD OUT** of **BPSK MODULATOR** to **DEMOD IN** of the **BPSKDEMODULATOR**.

7. Connect **DEMOD OUT** of **BPSK DEMODULATOR** to **DELAY IN** of **DELAY SECTION** and one input **DEPSKIN 1** of **DEPSK DECODER**.
8. Connect the **DELAY OUT** of **DELAY SECTION** to the input **DEPSK IN 2**of**DEPSK DECODER**.
9. Compare the DEPSK decoded data at **DEPSK OUT** with respect to input **SERIAL DATA.**
10. Observe various waveforms as mentioned below (Fig. 4.4)**.**

OBSERVATION:

Observe the following waveforms on oscilloscope and plot it on the paper.
ON KIT **DCL-BPSK**

1. **SERIAL DATA** with respect to **DATA CLOCK**. (Fig.4.4 (a))
2. Differentially encoded **DIFF OUT** with respect to **SERIAL DATA**. (Fig.4.4 (b))
3. Carrier frequency **SIN 0** and **SIN 180** (Fig.4.4 (c))
4. **MODOUT** with respect to **DIFF OUT**. (Fig.4.4 (d))
5. **MIDOUT** with respect to **DIFF OUT**. (Fig.4.4 (e))
6. **DEMODOUT** with respect to **DIFF OUT**. (Fig.4.4 (f))
7. **DELAYOUT** with respect to **DEMODOUT**. (Fig.4.4 (g))
8. **DEPSK OUT** with respect to **SERIAL DATA**. (Fig.4.4 (h))

CAPTURED WAVEFORM:

CH 1: DATA CLK (256 KHz) & CH 2: SERIAL DATA

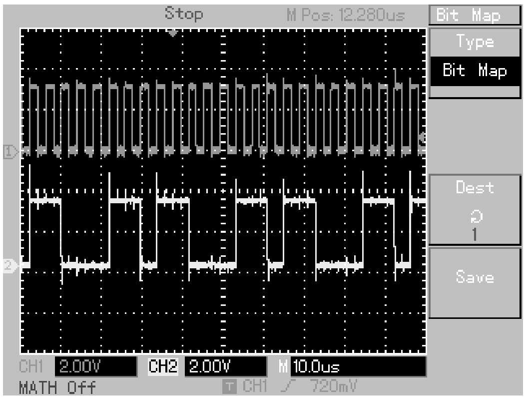

Fig.4.4 (a)

CH 1: SIN 0 (1 MHz) & CH 2: SIN 180 (1 MHz)

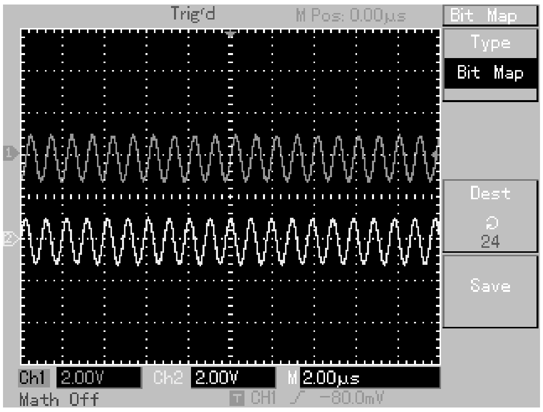

Fig.4.4 (b)

CH 1: SERIAL DATA& CH 2: DIFF OUT

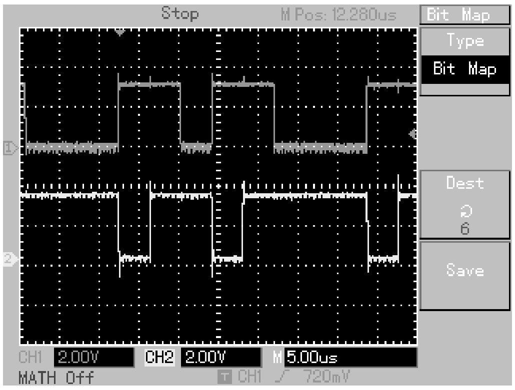

Fig.4.4(c)

CH 1: DIFF OUT& CH 2: MODOUT

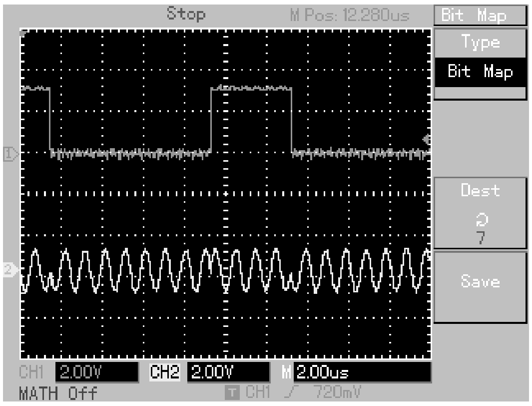

Fig.4.4 (d)

CH 1: DIFF OUT & MIDOUT

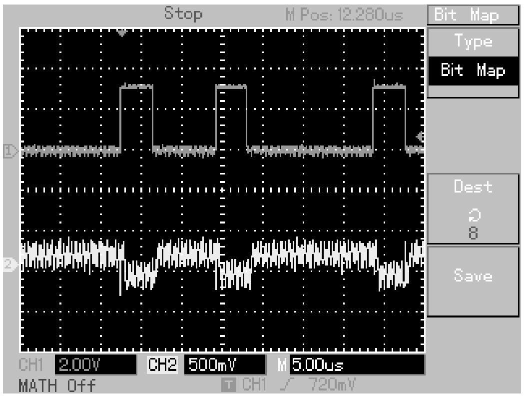

Fig.4.4 (e)

CH 1: DIFF OUT& CH 2: DEMODOUT

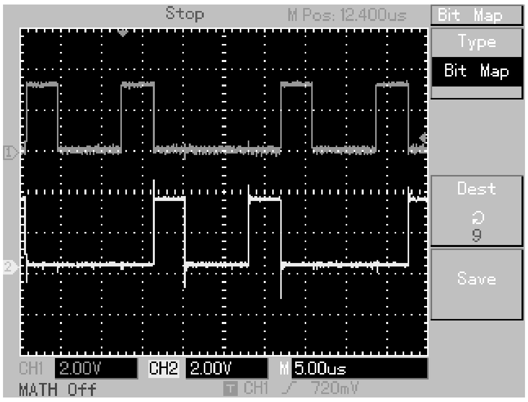

Fig.4.4 (f)

CH 1: SERIAL DATA. & CH 2: DEPSK OUT

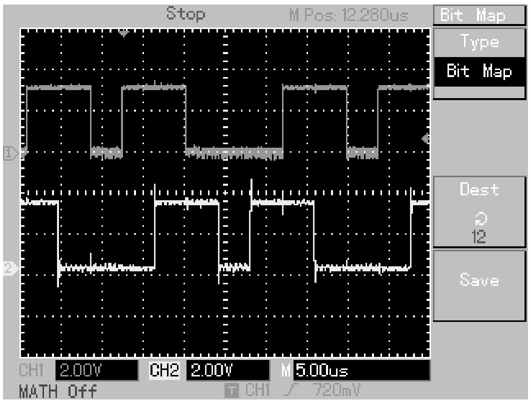

Fig.4.4 (h)

CONCLUSION:

The differential coding of data to be transmitted makes the bit “1” to be transformed into carrier phase variation. In this way the receiver recognizes one bit “1” at a time which detects a phase shift of the modulated carrier, independently from its absolute phase. In this way the BPSK modulation, which can take to the inversion of the demodulated data, is overcome. The only advantage of DEPSK over DPSK is that it is need not required complicated hardware to decode the differentially encoded data.